Progress Report: Smart Veterinary System with AI-Based Disease Detection

Course Code: CSE 299
Section: 04
Project Group: 05

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***Abstract*—This progress report summarizes the development work completed today for the project titled *Smart Veterinary System with AI-Based Disease Detection*. The team focused on improving and finalizing the shop module and appointment system of the platform. Tasks were distributed equally among team members, and each member contributed to both frontend and backend development using Django.**

I. Current Progress

During today's development session, the team worked on refining the shop system, implementing additional platform features, and improving system integration. Each member contributed equally to different modules of the project.

TABLE I
TODAY'S DEVELOPMENT TASKS

Completed Task	Responsible Member
Improved the shop interface, product display system, and integrated cart counter functionality	Rishat Bin Sultan
Developed and improved the appointment booking page for veterinary consultation	Nafia Haque Muskan
Implemented backend logic for appointment requests and managed database integration	MD Tamim Hasan
Performed system testing, UI improvements, and bug fixing for shop and appointment modules	Fariha Islam

II. Challenges and Decisions

Challenges:

- Maintaining synchronization between frontend interfaces and Django backend functionalities.
- Ensuring cart updates, appointment submissions, and database operations work correctly without conflicts.
- Organizing the overall system structure to prepare for AI integration in the next phase.

Decisions:

- The team decided to complete all remaining web system features before beginning AI model training.
- Each member continues to handle both frontend and backend aspects of assigned modules to ensure consistent development.

III. Future Plan

In the next phase of development, the team will focus on implementing the AI-based disease detection system.

- Collect and organize veterinary disease datasets related to pets and animals.
- Perform data preprocessing and dataset preparation for machine learning.
- Train disease detection models using Python and Scikit-learn.
- Evaluate model performance and optimize prediction accuracy.
- Integrate the trained AI model with the Django backend for real-time disease prediction.

IV. Conclusion

Today's development progress focused on improving the shop module and implementing the veterinary appointment system within the platform. The team successfully completed the assigned tasks through equal collaboration among members. With the web system now approaching completion, the next development stage will focus on dataset preparation, AI model training, and integration of the disease detection system.